higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

ROADS
NQF LEVEL 3

(12030363)

11 March 2019 (X-Paper)
09:00–12:00

This question paper consists of 6 pages.

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TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Write neatly and legibly.

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QUESTION 1: ROAD BUILDING TERMINOLOGY

1.1 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (1.1.1–1.1.10) in the ANSWER BOOK.

1.1.1 Asphalt is a mixture of water, cement and bitumen.

1.1.2 The spaces between the aggregate courses are filled with fine aggregate, mostly sand.

1.1.3 Texture treatment is the process used to restore the uniform texture of the road surface.

1.1.4 A layer of material less than 10 mm thick is excavated and removed from a road by a milling machine.

1.1.5 The objects and equipment installed on and next to roads and streets, usually for safety and information, are called road furniture.

1.1.6 Traffic signs are found along a road or street to give motorists important information.

1.1.7 Milling is the natural material on which the fill or pavement layer will be built.

1.1.8 Suitable markers used as references or guides during the initial excavation by the contractor surveyor are called grade lines.

1.1.9 A horizontal curve is a bend on the road leading to the left or to the right.

1.1.10 The gabion boxes are between 1 m × 4 m × 2 m wide.

(10 × 1) (10)

1.2 Define each of the following terms in road works:

1.2.1 Continuously graded asphalt

1.2.2 Directional sign

1.2.3 Borrow area

1.2.4 Camber or cross fall

1.2.5 Verge

(5 × 2) (10)
[20]

QUESTION 2: INTRODUCTION TO ROAD SURFACES

2.1 Name FIVE requirements to which the binder distributors must adhere. (5)

2.2 Give THREE types of equipment that can be used to construct the sand seal. (3)

2.3 Name the equipment used to test the laid asphalt. (1)

2.4 Briefly discuss the advantages of using concrete rather than asphalt on the road regarding each of the following:

2.4.1 Durability

2.4.2 Life span

2.4.3 Repair

2.4.4 Weather conditions

2.4.5 Damage

(5 × 1) (5)

2.5 Define the term *edge restraints.* (1)
[15]

QUESTION 3: ROAD FURNITURE

3.1 Name TWO different types of road signs and give an example of each. (2 × 2) (4)

3.2 Guidance signs are road furniture found on guiding vehicles on the roads.

Name THREE things with which the signs help motorists on the road. (3)

3.3 A speed hump is a device used to slow down traffic on the road and is considered as a traditional traffic calming solution.

Name THREE areas where speed humps can be installed. (3)
[10]

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QUESTION 4: ROAD SURVEY

4.1 Define each of the following terms used in road surveying:

4.1.1 Compound curve

4.1.2 Transition curve

4.1.3 Reverse curve

4.1.4 Simple curve

4.1.5 Centripetal force

(5 × 2) (10)

4.2 Give another name for a vertical curve. (1)

4.3 For what does each of the following abbreviations on vertical curves stand?

4.3.1 PVT

4.3.2 PVC

4.3.3 PVI

4.3.4 L

4.3.5 G1

4.3.6 G2

(6 × 1) (6)

4.4 Name TWO types of equipment that can be used to set out a crest curve. (2)

4.5 A theodolite is an instrument used for measuring angles.

Name SIX components of a theodolite. (6)

[25]

QUESTION 5: EARTHWORK

5.1 Define each of the following terms:

5.1.1 Compaction

5.1.2 California bearing ration

5.1.3 Relative density

5.1.4 Sand replacement test

5.1.5 Pycnometer

(5 × 2) (10)

5.2 Give ONE example of each of the following types of soil:

5.2.1 Cohesive soil

5.2.2 Noncohesive soil

(2 × 1) (2)

5.3 The modified AASHTO is a test used to calculate the maximum soil density achievable at different moisture contents.

State TWO functions of this test. (2)

5.4 The soil used to build a road must be tested before taken to the site to build the layers of the road.

Give FOUR different test used to tests the soil. (4)

5.5 For what does each abbreviation stand?

5.5.1 CBR

5.5.2 RD

(2 × 1) (2)

5.6 Briefly list FIVE steps to determine the dry density of soil by using the sand replacement test. (5)

5.7 Give FIVE steps to be carried out to remove material from the density hole. (5)

[30]

TOTAL: 100

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